

Muhammad Nadeem

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Professional Summary

AI Research Engineer specializing in **LLM/VLM and vision models**, with end-to-end ownership from research and training to optimized GPU inference. Builds **agentic RAG** systems (LangChain, tool calling, vector retrieval, evaluation), fine-tunes **lightweight LLM/VLMs** with PyTorch and Hugging Face (Transformers, PEFT, LoRA/QLoRA), and compresses them for **resource-constrained GPUs and edge devices** using quantization (INT8/4-bit), ONNX, TensorRT, OpenVINO, and vLLM/Ollama serving. Ships vision models against **live CCTV/IP-camera and drone RTSP streams**. M.S. in Computer Science (Sejong University, CVPR Lab); **13 publications** (5 first-author) on lightweight vision, vision-language, and edge inference — including WACV 2026 (LLVM-AD workshop) and ACPR 2025 — plus a Best Paper Award (ICNGC 2023). **Portfolio:** mnadeemdev.github.io

Education

- 2nd Mar-22 **Sejong University**, Seoul, South Korea
18th Feb-24
 - M.S. in Computer Science and Engineering
 - Thesis: FlameNet: An Efficient Deep Learning Framework for Fire Detection and Classification in Smart City Environment
 - GPA: 4.25/4.5, [Presentation Link](#)

1st Feb-14 **The Islamia University of Bahawalpur**, Bahawalpur, Pakistan
23rd Apr-18
 - B.S. in Computer Science
 - GPA: 3.28/4.0

Technical Skills

LLM / VLM	Qwen2.5-VL, HyperCLOVA X SEED, CLIP-style dual encoders, Hugging Face Transformers & PEFT, LoRA/QLoRA fine-tuning, instruction tuning, chain-of-thought prompting, zero-shot and prompt-driven cross-modal recognition, VQA, vLLM and Ollama serving, GGUF conversion (llama.cpp)
RAG / Retrieval	Agentic RAG, LangChain, hybrid retrieval (dense + BM25 + knowledge graph), BGE-M3 embeddings, Reciprocal Rank Fusion, cross-encoder reranking, Personalized PageRank multi-hop search, vector databases (Chroma), citation grounding, RAG evaluation
Inference Opt.	Quantization (INT8 / 4-bit, PTQ & QAT), ONNX / ONNX Runtime, TensorRT (FP16/INT8 engines), OpenVINO, TFLite, knowledge distillation, pruning, batching & KV-cache tuning, latency/VRAM profiling
Vision Models	YOLOv8/v12/v26, Faster R-CNN, DETR, SSD, YOLOX, YOLO-NAS, GroundingDINO, Mask R-CNN instance segmentation, MobileNet and lightweight CNN design, tiny/small-object detection
Tracking / Pose	Deep SORT multi-object tracking, YOLO-pose 17-keypoint skeleton estimation, activity and posture recognition, fall/immobility detection, face de-identification (blur, pixelate, solid masking) with audit logging
Generative	FLUX.2 [dev], Stable Diffusion, GANs, Hugging Face diffusers, image-conditioned generation, synthetic data generation for data-scarce detection tasks
Frameworks	PyTorch, Hugging Face, TensorFlow, OpenCV, Caffe, CUDA-based multi-GPU training (DP, DDP, multi-node)
Serving	FastAPI, Flask, REST APIs, real-time inference on live RTSP streams (IP/CCTV cameras, drone feeds), OpenCV video ingestion, React dashboards
GPU / Edge HW	NVIDIA GPU servers, Jetson Orin Nano, Raspberry Pi 4/5
Time-Series	N-BEATS, N-HiTS, forecasting and prediction pipelines
Lang & Tools	Python, C++, Java, C#, Git, JIRA, Agile/Scrum, VS Code, PyCharm, Jupyter; Ubuntu 24.04, Windows 11, macOS

Selected Projects (Portfolio)

Full portfolio with demos and code: mnadeemdev.github.io | github.com/MNadeemDev

- 2026 **GaN Fab Process-Traceability RAG System** (on-premise LLM + VLM)
Stack: HyperCLOVA X SEED, Qwen2.5-VL 7B, BGE-M3, bge-reranker-v2-m3, Hugging Face PEFT (LoRA/QLoRA), Ollama, llama.cpp/GGUF, PP-StructureV3
- Built a fully **on-premise Korean-language RAG assistant** over a 14,024-chunk knowledge base combining 12,890 equipment trouble/maintenance records with a 314-page MES manual.
 - Designed **three-signal hybrid retrieval** — BGE-M3 dense vectors, BM25 with a Korean syllable-bigram tokenizer, and a multi-hop knowledge graph traversed by Personalized PageRank — fused with Reciprocal Rank Fusion, then reranked top-30→top-5 by a cross-encoder, returning every answer with citations.
 - Used **Qwen2.5-VL 7B** to describe manual screenshots (menu trees, buttons, highlighted regions, table columns), making content that exists only inside images retrievable as text.
 - Domain-adapted the generator with **LoRA and 4-bit QLoRA** via Hugging Face PEFT; merged adapters and converted to GGUF (llama.cpp) for Ollama serving.
 - Engineered a 5-step chain-of-thought root-cause-analysis prompt and automated traceability/RCA/maintenance report generation with auto-generated citations.
 - Benchmarked candidate LLMs on an in-house RCA benchmark (cause accuracy, numeric fidelity, citation compliance, Korean fluency) to select the deployed model.
- 2025 **SpectroFire — Zero-Shot EO/IR Vision-Language Recognition** (ACPR 2025, ICOIN 2026)
- 2026 **Stack:** CLIP-style dual encoder, contrastive learning, prompt engineering, EO/IR fusion, PyTorch, Jetson
- Built a CLIP-style dual-encoder over **EO + thermal-IR** imagery for zero-shot recognition, classifying by cosine similarity against prompt embeddings so new target classes are added by writing a prompt — no retraining.
 - Added cross-spectral registration, temporal synchronization, density-based geolocation, and structured event messaging, with automatic CUDA/CPU device selection for Jetson deployment.
 - Combined real-time edge inference with IoT data networking for distributed monitoring (ICOIN 2026).
- 2026 **PA-Drive — End-to-End Autonomous Driving Planner** (2026 Autonomous Driving AI Challenge)
- Stack:** Transformer decoder, VAD/VADv2 scene encoder, nuScenes, FP16 autocast, Docker
- Built a 6-camera end-to-end planner: scene encoder → K trajectory-query candidates from a Transformer decoder → differentiable physical objective (collision, lane, dynamics, comfort, progress) → hard-constraint filtering → ranking.
 - Cut planning L2 error from **1.902 to 0.246** on nuScenes across four iterations by relativizing the confidence gate, adding ego-conditioned kinematic anchors, and correcting three constraint definitions.
 - Eliminated temporal data leakage by moving from random frame splits to official nuScenes splits (leaked 0.485 vs. true 1.289), and enforced FP16 autocast with an explicit FLOPs budget check for on-vehicle latency.
 - Related first-author paper on lightweight multi-scale fusion for real-time driving segmentation accepted at **WACV 2026 (LLVM-AD workshop)**.

- 2026 **Manhole Worker Safety — Pose-Based Activity Tracking and Risk Fusion** (ICRA 2027 submission)
Stack: YOLOv8/YOLO26-pose (Ultralytics), 17-keypoint skeleton tracking, OpenCV, RTSP, multi-sensor fusion, ONNX Runtime, OpenVINO, FastAPI, Intel N100 edge PC
- Built **worker activity tracking with pose estimation**: per-person detection plus 17-keypoint skeletons tracked across frames from live RTSP/IP-camera feeds to monitor posture and activity inside confined spaces.
 - Implemented **fall and immobility detection** from tracked pose geometry — torso tilt beyond 55° from vertical or bounding-box aspect above 1.10, confirmed over 5 consecutive frames to suppress flicker — alongside an fps-normalized stillness detector flagging prolonged immobility.
 - Designed **dual-track de-identification** so a single pose model serves privacy and safety simultaneously: a monitoring track blurs or pixelates faces while keeping the body skeleton visible for fall judgment, and a storage track applies an irreversible solid mask. The face region is derived from facial keypoints with a shoulder-based fallback so no face is left unmasked in low light or backlit conditions.
 - Emitted a per-frame audit log (persons detected, faces masked, fallback usage, fall alarms) as compliance evidence for de-identification requirements.
 - Fused this camera-based worker state with a five-measurand gas array into a **four-level risk lattice** driving distinct response protocols, with every hazard decision made **connectivity-independent at the edge node** for underground sites without reliable wireless.
 - Benchmarked ONNX Runtime and OpenVINO inference latency (mean/p50/p90/p99) on an NPU-less 4-core Intel N100 board, and delivered the system with FastAPI services and an operator dashboard (live grid, map, history, mobile views, i18n).
- 2026 **Drone-Based Safety-Helmet Compliance Detection with Generative Data Synthesis**
Stack: FLUX.2 [dev] via Hugging Face diffusers, YOLO, PyTorch, bfloat16, model CPU offload
- Built aerial detection of **safety-helmet non-compliance** for construction sites and ski resorts, flagging workers operating without helmets from drone imagery.
 - Addressed severe **data scarcity** — real footage of helmet violations is rare and cannot be staged safely — by synthesizing training images with **FLUX.2 [dev]** image-conditioned generation from a reference photo.
 - Programmatically randomized prompts across backgrounds (construction site, factory yard, bridge, warehouse), weather and lighting, worker positions, and machinery to build a diverse 1024×1024 dataset at a fixed ~10 m aerial viewpoint.
 - Ran the diffusion model in **bfloat16 with model CPU offload**, streaming weights between host RAM and GPU so generation fit on consumer-grade cards.
 - Augmented a real annotated helmet-detection dataset (YOLO format) with the synthetic imagery to train and evaluate the detector.
- 2024 **Real-Time Detection and Tracking on CCTV and Drone Streams**
Present **Stack:** YOLOv8/v12/v26, Deep SORT, RTSP/OpenCV ingestion, TensorRT, FastAPI, Jetson Orin Nano
- Built inference pipelines consuming **live RTSP streams from IP/CCTV cameras and drone feeds**, handling decoding, frame sampling, and reconnection for continuous operation.
 - Delivered detection and multi-object tracking on these streams: foreign object debris, sea-water waste, fire and smoke, and person/object tracking.
 - Tuned tiny/small-object detection and optimized models with TensorRT for real-time throughput on GPU and edge hardware.
 - Served models as REST APIs with visualization dashboards for operators.
- 2022 **FlameNet — Lightweight Fire Detection for Smart Cities** (M.S. Thesis, published)
- 2024 **Stack:** PyTorch, CNN design, model compression, IoT/edge deployment
- Designed an efficient deep learning architecture for fire detection and classification in IoT environments, prioritizing low parameter count and inference cost.
 - Published in *MDPI Smart Cities* (IF 7.0, Q1); related lightweight CNN work received the **Best Paper Award** at ICNGC 2023.
- 2022 **Fake QR Code Recognition with a Lightweight MobileNet Classifier**
Stack: MobileNet, PyTorch, transfer learning, lightweight CNN design
- Built an image classifier that distinguishes **authentic from forged QR codes**, targeting counterfeit and tampered-code detection.
 - Selected **MobileNet** for its low parameter count and depthwise-separable convolutions so verification could run on phones and low-power devices rather than a server.
 - Trained by transfer learning and evaluated genuine-vs-fake separation to tune the accuracy/model-size trade-off.

- 2022 **VPBR — Pumpkin Biophysical Property Measurement with Mask R-CNN** (published)
- 2023 **Stack:** Mask R-CNN instance segmentation, PyTorch, OpenCV, geometric measurement
- Built an automatic, low-cost vision pipeline that segments individual pumpkins with **Mask R-CNN** and converts the instance masks into **length and biophysical property measurements**, replacing manual measurement in the field.
 - Handled instance separation under occlusion and varied viewpoints so per-fruit geometry could be derived reliably from the predicted masks.
 - Published in *MDPI Plants* (IF 4.5, Q1) as *VPBR: An Automatic and Low-Cost Vision-Based Biophysical Properties Recognition Pipeline for Pumpkin*.

Selected Publications

13 publications; 5 as first author. Full list: [Google Scholar](#)

Vision-Language Models, Lightweight Models and Edge Inference (2025–2026)

- WACV 2026 **Lightweight Multi-Scale Fusion for Real-Time Autonomous Driving Segmentation**
Workshop Muhammad Nadeem, Hyeyoung Lee
[\[IEEE/CVF WACV 2026 Workshops — 5th Workshop on Large Language and Vision Models for Autonomous Driving \(LLVM-AD\), pp. 1721–1728\]](#)
- ACPR 2025 **Bringing CLIP to the Edge: A Lightweight Fire Detection System for Real-Time Monitoring with UAV**
 Hyeyoung Lee, Muhammad Nadeem
[\[Asian Conference on Pattern Recognition \(ACPR\), 2025\]](#)
- ICRCV 2025 **A Label-Free Lightweight Prompt-Driven Cross-Modal Fire Detection on Robotic Edge Platforms**
 Hyeyoung Lee, Muhammad Nadeem
[\[The 7th International Conference on Robotics and Computer Vision \(ICRCV\), 2025\]](#)
- ICOIN 2026 **SpectroFire: Zero-Shot Fire and Smoke Detection via Real-Time Edge Inference and IoT Data Networking**
 Hyeyoung Lee, Muhammad Nadeem
[\[The 40th International Conference on Information Networking \(ICOIN\), 2026\]](#)
- ICARCA 2025 **EdgeFlame: A Physics-Inspired Framework for Zero-Shot Fire and Smoke Recognition**
 Hyeyoung Lee, Muhammad Nadeem
[\[The 4th International Conference on Automation, Robotics and Computer Applications \(ICARCA\), 2025\]](#)

Applied AI, IoT and Signal Processing

- BigComp 2025 **IoT Anomaly Detection with Sound-Based Spectral Analysis and Lightweight Model**
 Hyeyoung Lee, Muhammad Nadeem
[\[IEEE International Conference on Big Data and Smart Computing \(BigComp\), 2025\]](#)
- BigComp 2026 **Physics-Consistent Hybrid AI for Quantum Device Manufacturing**
 Hyeyoung Lee, Muhammad Nadeem
[\[IEEE International Conference on Big Data and Smart Computing \(BigComp\), 2026\]](#)
- arXiv 2025 **Spectral Feature Extraction for Robust Network Intrusion Detection Using MFCCs**
 Hyeyoung Lee, Muhammad Nadeem, Piotr Tsoi
[\[arXiv preprint, 2025\]](#)

Vision and Deep Learning (M.S. Research, 2023)

- Smart Cities **Visual Intelligence in Smart Cities: A Lightweight Deep Learning Model for Fire Detection in an IoT Environment**
IF 7.0, Q1 Muhammad Nadeem, Naqqash Dilshad, Norah Saleh Alghamdi, L Minh Dang, Hyoung-Kyu Song, Junyoung Nam, Hyeonjoon Moon
[\[MDPI, PDF\]](#)
- Plants **VPBR: An Automatic and Low-Cost Vision-Based Biophysical Properties Recognition Pipeline for Pumpkin**
IF 4.5, Q1 L. Minh Dang, Muhammad Nadeem, Tan N. Nguyen, Han Yong Park, O New Lee, Hyoung-Kyu Song, Hyeonjoon Moon
[\[MDPI, PDF\]](#)

- ITCCSCC-2023 **Machine Learning for Mental Health: A Systematic Study of Seven Approaches for Detecting Mental Disorders**
Muhammad Nadeem, Junaid Rashid, Hyeonjoon Moon, Arailym Dosset
[\[The 38th International Technical Conference on Circuits/Systems, Computers and Communications, PDF\]](#)
- ICTC-2023 **Wheat Diseases Recognition Using Optimal Features Assisted Modified Soft Attention Network**
Muhammad Nadeem, Aqib Khan, Ji-Won Kim, L. Minh Dang, Hyeonjoon Moon
[\[The 15th International Conference on ICT Convergence\]](#)
- ICNGC-2023 **Brest Cancer Recognition Through Visual Intelligence Assisted Lightweight Convolution Neural Network**
Best Paper Award
Muhammad Nadeem, Haseeb Khan, Wisal Khan, L. Minh Dang, Nguyen Le Quan, Hyeonjoon Moon
[\[The 9th International Conference on Next Generation Computing \(ICNGC 2023\)\]](#)

Research and Work Experience

- 15th April-2024 **AI Research Engineer** (LLM/VLM & Computer Vision), Spilab, Seoul, South Korea
 Present
- Tools:** LLM/VLM (Hugging Face, PEFT/LoRA, LangChain, agentic RAG, vLLM), Optimization (quantization, ONNX, TensorRT, OpenVINO, TFLite), CV (YOLOv8/12/26, Deep SORT), Serving (FastAPI, Flask)
- Research and development of vision models and algorithms for detection, segmentation, and tracking on live CCTV/IP-camera and drone video streams.
 - Developed agentic RAG and LLM technologies: an on-premise hybrid-retrieval assistant (dense + BM25 + knowledge-graph fusion, cross-encoder reranking, cited answers) over a 14,000-chunk industrial knowledge base, plus a production multilingual AI chatbot.
 - Fine-tuned lightweight LLM/VLMs with Hugging Face Transformers and PEFT (LoRA/QLoRA) under constrained GPU memory, then merged adapters and converted to GGUF for local serving.
 - Optimized inference through quantization (INT8/4-bit), ONNX, TensorRT, OpenVINO, and vLLM/Ollama serving, cutting latency and VRAM for single-GPU and edge deployment.
 - Applied VLMs to document understanding, using Qwen2.5-VL to make text and UI content inside manual screenshots retrievable, alongside OCR and VQA pipelines.
 - Delivered applied CV projects: foreign object debris detection, sea-water waste detection, drone-based fire and smoke detection, safety-helmet compliance monitoring, and person/object tracking.
 - Built pose-estimation-based worker activity tracking with fall detection and privacy-preserving face de-identification for confined-space safety monitoring.
 - Generated synthetic training data with diffusion models (FLUX.2 [dev], Stable Diffusion) to overcome data scarcity on safety-critical detection tasks where real examples are rare.
 - Deployed models to edge hardware (Jetson Orin Nano, Raspberry Pi) with OpenVINO/TFLite conversion for resource-constrained inference.
 - Developed time-series forecasting with N-BEATS and N-HiTS, and speech emotion recognition models.
 - Published 8 papers during this role (2025–2026) on vision-language and lightweight edge models, including WACV 2026 workshop (LLVM-AD), ACPR 2025, ICRCV 2025, and ICOIN 2026; also present work at conferences and write technical reports.

- 15th Mar-22 **Graduate Researcher** (Computer Vision), CVPR Lab, Sejong University, Seoul, South Korea
 28th Feb-24 **Tools:** PyTorch, Mask R-CNN, YOLO/Faster R-CNN, Stable Diffusion, multi-GPU training, dataset development, RNNs, GANs
- University AI research lab: designed and evaluated vision models for image classification, segmentation, and object detection, producing 5 peer-reviewed publications.
 - Designed lightweight/efficient CNN architectures (including MobileNet-based classifiers for fake QR code recognition) for fire detection, medical imaging, and security applications, optimizing accuracy against parameter count and inference cost.
 - Built Mask R-CNN instance-segmentation pipelines converting predicted masks into physical measurements for agricultural phenotyping (published in *MDPI Plants*).
 - Scaled training with Data Parallel, Distributed Data-Parallel, and multi-node setups on GPU clusters.
 - Training, fine-tuning CV models, transfer learning, and evaluating their performance.
 - Built and curated datasets, and applied generative models (Stable Diffusion, GANs) for data augmentation.
 - Write research papers, and technical reports, and present the work at conferences and meetings.
- 1st Jul-2018 **Software Engineer**, Systems Junction, Lahore, Pakistan
 30th June-2021 **Tools:** Android (Java, Android Studio), Networking (Retrofit), Storage (Room), Media (Glide/Picasso), Backend (Firebase)
- Designing, developing, and maintaining Android applications using Java and XML programming languages.
 - Writing clean, efficient, and maintainable code that follows best practices and coding standards.
 - Troubleshooting and debugging issues, and implementing fixes to ensure smooth functioning.
 - Providing technical guidance and mentorship to junior team members, as needed.
 - Work on technical documentation, such as architectural diagrams, code, and user manuals.
 - Worked on MVC and singleton pattern, real-time push notifications, Firebase, realm
 - Optimizing computer vision algorithms for performance and memory usage on mobile devices.
 - Maintaining and improving existing CV codebases, including refactoring and bug fixing.
 - Developing and maintaining image processing pipelines and data processing workflows.

Honors and Achievements

- 2023 **Best Paper Award**, ICNGC 2023
 2022 **Full Tuition Fee Waiver**, School of Computer Engineering, Sejong University
 2022 **Prof. Stipend**, CVPR Lab, Sejong University
 2023 **Conference Travel Grant**, Ministry of Education, South Korea
 2014 **Merit Scholarship for Bachelor Degree**, The Islamia University of Bahawalpur, Pakistan

Academic Service

- Journal** Reviewer: IEEE Trans. on Pattern Analysis and Machine Intelligence (TPAMI), IEEE Trans. on Image Processing (TIP), MDPI Fire
- Conference** Reviewer: 10th Int. Conf. on Next Generation Computing; 40th ITC-CSCC

English Proficiency and Relevant Courseworks

- IELTS-AC 6.0** Listening 6.0, Speaking 6.0, Writing 6.0, Reading 5.5
- Languages** English (4/5), Korean (1/5)
- A+** Introduction to Deep Learning, Advanced Deep Learning, Advanced AI
- A+** Computer Vision Topics
- A0** Advanced HCI

References

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